

BROOKS COUNTY AP, TX, USA

WMO: 720269

Lat: 27.207N

Lon: 98.121W

Elev: 112

StdP: 14.64

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12982

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.6	37.2	16.2	12.5	54.8	20.4	15.4	55.3	24.5	75.2	21.1	71.8	5.1	350	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 23.1	(c) 101.7	(d) 75.2	(e) 99.7	(f) 75.4	(g) 98.0	(h) 75.4	(i) 80.8	(j) 89.7	(k) 79.9	(l) 89.0	(m) 79.2	(n) 88.3	(o) 11.5	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.0	150.9	83.5	77.7	144.6	82.7	77.1	141.9	82.5	44.2	89.7	43.4	88.8	42.7	87.8	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	18.6	16.8	DB	30.8	105.5	2.8	3.2	28.8	107.7	27.2	109.6	25.6	111.4	23.6	113.7
			WB	27.0	81.9	2.9	1.3	25.0	82.9	23.3	83.6	21.7	84.4	19.6	85.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.2	58.6	63.8	69.3	75.4	81.0	84.9	85.2	86.0	82.4	76.0	67.0	60.6
	DBStd	11.96	9.30	10.04	8.47	6.71	5.28	3.20	4.12	4.22	3.88	6.81	9.02	9.53
	HDD50	68	29	13	3	0	0	0	0	0	0	0	5	18
	HDD65	735	239	133	57	8	0	0	0	0	0	9	86	203
	CDD50	8910	297	398	602	762	962	1048	1091	1115	972	805	513	345
	CDD65	4102	42	98	191	319	497	598	626	650	522	349	144	66
	CDH74	44655	373	765	1514	3024	5327	7168	7740	8145	5440	3433	1215	511
	CDH80	21986	72	241	539	1314	2570	3815	4211	4574	2654	1505	377	114
Wind	WSAvg	7.2	6.8	7.6	8.2	8.7	8.7	8.0	7.8	6.8	5.3	6.0	6.4	6.6
Precipitation	PrecAvg	25.9	1.2	1.5	0.9	1.3	3.0	3.1	1.8	2.7	4.4	3.1	1.3	1.2
	PrecMax	55.2	3.8	6.2	4.7	5.4	12.4	13.0	11.6	12.9	32.8	10.4	5.2	5.0
	PrecMin	12.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0	0.0
	PrecStd	9.1	0.9	1.4	0.9	1.4	2.4	3.0	2.2	3.0	5.0	2.7	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	91.1	94.9	98.9	100.4	102.5	103.9	104.9	100.4	96.3	90.8	87.2
		MCWB	66.3	68.7	68.1	73.3	74.6	74.9	74.1	76.4	73.1	75.2	72.7	69.5
	2%	DB	81.7	86.4	89.5	94.8	97.8	99.4	100.8	101.9	97.5	93.0	87.9	83.0
		MCWB	65.4	67.5	70.0	71.6	74.5	75.5	75.1	76.0	74.4	74.9	71.7	67.7
	5%	DB	78.6	82.7	86.0	90.7	95.1	97.3	98.8	99.7	95.1	90.7	84.4	79.9
		MCWB	63.7	66.9	68.3	70.7	74.8	75.5	75.7	76.2	75.9	73.9	70.0	66.1
	10%	DB	74.4	79.0	82.2	87.9	91.7	95.2	96.0	97.4	92.1	88.4	81.3	76.3
		MCWB	62.7	65.0	67.5	70.6	73.9	75.9	76.1	76.4	75.8	73.5	68.4	66.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.6	72.4	74.0	78.4	80.2	81.3	81.6	82.4	81.1	80.0	76.2	73.1
		MCDB	77.5	82.8	84.7	89.7	92.3	92.4	89.0	92.5	89.1	87.8	84.5	79.3
	2%	WB	69.5	71.1	72.5	76.4	78.5	79.8	80.2	80.8	79.9	78.4	74.7	71.4
		MCDB	75.6	79.6	81.6	86.6	89.0	90.1	88.7	89.8	88.0	86.5	82.3	77.7
	5%	WB	67.9	69.9	71.4	75.0	77.7	78.9	79.2	79.7	78.9	77.4	73.1	70.2
		MCDB	75.1	77.5	79.4	84.5	87.2	88.6	88.2	88.6	87.1	85.1	80.1	76.6
	10%	WB	65.9	68.7	70.5	73.8	76.9	78.2	78.4	78.9	78.1	76.3	71.6	68.5
		MCDB	72.6	76.0	77.9	82.3	85.9	87.5	87.3	88.0	86.4	84.0	78.6	74.6
Mean Daily Temperature Range	5% DB	MDBR	22.2	22.0	22.0	22.9	21.3	21.2	21.9	23.1	20.1	22.6	22.4	21.5
		MCDBR	27.3	25.4	24.6	25.4	24.3	24.0	24.8	25.8	23.7	22.9	23.6	25.5
		MCWBR	13.8	10.4	9.2	8.9	6.3	5.6	5.5	5.8	6.6	7.9	9.9	12.8
	5% WB	MCDBR	20.7	20.5	19.9	20.4	19.4	20.2	20.4	22.1	19.0	19.5	19.7	20.3
MCWBR		12.2	9.3	8.6	7.8	6.1	5.8	5.9	6.0	6.3	8.3	9.8	12.5	
Clear-Sky Solar Irradiance	taub		0.354	0.368	0.385	0.422	0.455	0.438	0.437	0.437	0.427	0.379	0.367	0.348
	taud		2.453	2.408	2.369	2.280	2.224	2.329	2.342	2.349	2.344	2.470	2.448	2.482
	Ebn at Noon		283	287	288	279	268	271	271	271	271	280	275	279
	Edh at Noon		31	35	38	43	46	41	40	40	39	32	31	29
All-Sky Solar Radiation	RadAvg		985	1182	1466	1740	1895	2116	2102	2020	1662	1443	1098	884
	RadStd		110	197	177	170	160	93	161	105	143	159	120	130

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	N/A	+232	+211

Nomenclature: See separate page